

E7 Introduction to Computer Programming for Scientists and Engineers

Discussion week 11

Numerical differentiation and integration

Recap

Numerical differentiation

Name	Approximates	Formula	Error bound
Forward difference	$f'(x)$	$\frac{f(x+h) - f(x)}{h}$	$\frac{M_2}{2}h$
Backward difference	$f'(x)$	$\frac{f(x) - f(x-h)}{h}$	$\frac{M_2}{2}h$
Central difference	$f'(x)$	$\frac{f(x+h) - f(x-h)}{2h}$	$\frac{M_4}{6}h^2$
Central difference	$f''(x)$	$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$	$\frac{M_4}{12}h^2$

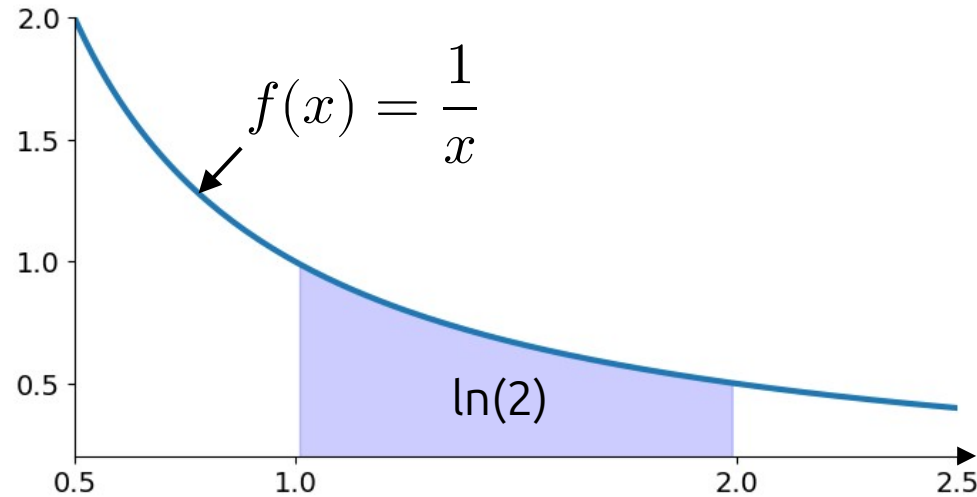
Numerical integration

All approximate $\int_a^b f(x) dx$

Name	Formula	Error bound
Trapezoidal rule	$\frac{h}{2}(f_0 + f_1)$	$\frac{M_2}{12}h^3$
Composite trapezoidal rule	$\frac{h}{2} \sum_{i=1}^n (f_{i-1} + f_i)$	$\frac{(b-a)M_2}{12}h^2$
Simpson's 1/3 rule	$\frac{h}{3}(f_0 + 4f_1 + f_2)$	$\frac{M_4}{90}h^5$
Composite 1/3 rule	$\frac{h}{3} \sum_{i=1}^{n/2} (f_{2i-2} + 4f_{2i-1} + f_{2i})$	$\frac{(b-a)M_4}{180}h^4$

Example 1

Knowing that $\int_1^2 \frac{1}{x} dx = \ln(2)$, what is the minimum number of grid points needed to compute $\ln(2)$ with an error of at most 0.01, using either composite trapezoidal or composite Simpson?



Example 2

Knowing that $\int_0^1 \frac{1}{1+x^2} dx = \frac{\pi}{4}$, what is the minimum number of grid points needed to compute π with an error of at most 0.01, using either composite trapezoidal or composite Simpson?

